Highlights

Beyond Directional Accuracy: A Verified Walk-Forward Framework for Machine-Learning Cryptocurrency Trading

- A tested walk-forward framework for daily crypto forecasting without leakage
- Logistic forecasts beat a coin flip on BTC, ETH and DOGE (52.95% pooled)
- Size-aware tests show where the model is better than a one-day reversal rule
- A joint return and turnover bootstrap puts the break-even cost at 11.4 bps
- Mutation tests confirm the engine's safeguards: 8 of 8 violations caught

Beyond Directional Accuracy: A Verified Walk-Forward Framework for Machine-Learning Cryptocurrency Trading

Abstract

Most studies judge machine-learning forecasts of cryptocurrency returns by directional accuracy alone. Accuracy does not show whether a model adds anything to a simple rule, or whether it still earns money after trading costs. We build an evaluation framework that links forecasting, statistical testing and trading economics, avoids look-ahead leakage and tests its own code, and we apply it to daily Binance spot data for Bitcoin, Ether and Dogecoin. The framework uses 30 causal features, walk-forward validation with in-fold scaling and a training–test gap, paired tests against simple baselines, measures based on the size of price moves, and a break-even cost estimated together with turnover. Its tests caught all eight rule violations that we put back into the backtest engine. Over 5,562 out-of-sample forecasts from July 2020 to March 2026, a regularized logistic model is correct 52.95% of the time (95% interval 51.45–54.41%) and beats a coin flip on every asset. A one-day reversal rule scores 52.88%, so much of this skill is shared with it, but the model is better on large moves. It has the higher return-weighted hit rate on all three assets, and its long/short book earns a gross Sharpe ratio of 0.54, against 0.004 for the reversal rule. At 9 basis points per unit of turnover the net Sharpe ratio is 0.11 and the break-even cost is 11.4 basis points, while a long-or-cash version reaches 0.71. The bootstrap intervals are still wide. The full analysis is reproduced by one command.

Keywords: Financial machine learning, Walk-forward validation, Cryptocurrency, Forecast evaluation, Transaction costs, Backtest verification
JEL: G17, C53, C58, G14

1. Introduction

Machine-learning models are now a common tool for forecasting cryptocurrency prices. Most studies report their success as directional accuracy, which is the share of days on which the next move is predicted correctly. Accuracy is easy to compute and easy to compare, but it leaves three practical questions

open. Does the model know anything that a simple rule does not know? Does it get the right days correct, that is, the days with large moves that decide most of the profit and loss? And does its edge survive the cost of trading on it? A good model alone cannot answer these questions. We also need an evaluation process in which forecasting, testing and trading economics are linked, and in which the code behind each number can be trusted.

In this paper we build such a process and apply it to daily Binance spot data for Bitcoin (BTC), Ether (ETH) and Dogecoin (DOGE) from 2017 to 2026. The framework has five layers and a verification layer under them (Figure 1). The data layer checks the input files and records their hashes. The feature layer builds 30 strictly causal predictors, and it screens any feature that is supposed to come from outside the market for hidden dependence on prices. The learning layer fits the models in an expanding walk-forward design. Preprocessing is fitted inside each fold, and there is a gap between the training data and the test data. The inference layer compares every model with simple baselines on the same dates. It uses paired tests that allow for dependence, a bootstrap that resamples dates, and measures that take the size of moves into account. The economic layer turns the forecasts into a portfolio, estimates the break-even trading cost together with turnover, and checks the result against a separate cash ledger. The verification layer writes the timing and accounting rules of the backtest engine as executable contracts, and it uses mutation testing to confirm that the tests really catch violations.

We apply the framework to 5,562 out-of-sample forecasts made between July 2020 and March 2026, and we find four main results. First, an ℓ_2-regularized logistic regression is correct 52.95% of the time. The date-synchronized 95% interval for this pooled accuracy is 51.45–54.41%, and the model beats a coin flip on every asset. The Pesaran–Timmermann test confirms that its calls are related to the realized direction (p-values from 0.008 to 0.016), and the model is more accurate than gradient-boosted trees on all three assets. Second, the comparison with baselines shows that much of this skill is shared with one-day reversal. A rule that simply bets against the previous day's move is correct 52.88% of the time, and the model agrees with it on 71% of asset-days. Third, the model is different from that rule in magnitude. On the days where the two disagree, the model tends to be the one that gets the large moves right. It has the higher return-weighted hit rate on all three assets, and an equal-weight long/short book built on its forecasts earns a gross Sharpe ratio of 0.54, against 0.004 for the same book built on the reversal rule. Fourth, the economic result is positive at realistic costs. At 9 basis points (bps) per unit of turnover the net Sharpe ratio is 0.11, the estimated break-even cost is 11.4 bps, and a long-or-cash version reaches 0.71. With less than six years of daily data the bootstrap intervals

around these numbers are wide, and we report them everywhere.

Two more results are about the framework itself. Its provenance screen flags a synthetic sentiment series as price-derived on all three assets. This series was built from daily prices as a placeholder during development, and on one of the assets a linear check would have missed it. The verification layer catches all eight contract violations that we put back into the engine. The full pipeline runs again from one command, and in a clean environment with the pinned library versions it gives the same primary output files byte for byte.

The paper makes five contributions.

1. **An end-to-end evaluation algorithm** (Algorithm 1). It links forecasting, testing and trading economics, stores the date of every forecast, saves every fitted pipeline and produces every reported number from one command.
2. **A leakage-safe design**. Features are causal and are checked by a prefix-invariance test, preprocessing is fitted inside each fold, training and test data are separated by a gap, and a rank-based provenance screen checks external features.
3. **Inference that uses baselines and magnitude**. We use paired HAC comparisons against simple rules with no parameters, a bootstrap that resamples dates for pooled statistics, and return-weighted measures that show where the value of a model comes from.
4. **Cost-aware economics**. We estimate a break-even cost together with turnover, and we report a full cost curve, policy variants and a cash ledger that is reconciled separately.
5. **A verified implementation**. Ten executable contracts cover timing, accounting, statistics and estimation, and eight of them are confirmed by mutation testing.

We use simple, well-known models on purpose. Our aim is a process that tells a researcher what a model adds and whether it can pay for its trading costs. Any classifier can be used in the learning layer.

2. Related work

2.1. Backtest overfitting and data snooping

Most work on the reliability of backtests is about inference, that is, how much an attractive result should be discounted for the number of alternatives that were tried. White (2000) proposed the Reality Check bootstrap. Sullivan et al. (1999) applied it to thousands of technical trading rules on a century

of Dow Jones data, and they showed that the apparent significance of the best rule depends strongly on taking the whole set of searched rules into account. Hansen (2005) improved the procedure into a test for superior predictive ability. Bailey and López de Prado (2014) deflate the Sharpe ratio for selection and non-normality. Harvey et al. (2016) argue that the usual $t > 2$ hurdle is far too low, given how many factors have been tested, and Hou et al. (2020) find that most of several hundred published anomalies do not replicate under consistent methods. López de Prado (2018) gives the purging and embargo tools that we build on, and Arnott et al. (2019) set out a protocol for machine-learning backtests.

These studies usually take the backtest engine as given and ask what can be concluded from its output. Our framework adds a layer below this, in which the engine itself is checked. Errors in backtest code tend to have large effects, not small ones, and a natural tool for finding them is mutation testing. Mutation testing is standard in software engineering (Jia and Harman, 2011), but as far as we know it is rarely reported in empirical finance.

2.2. Benchmarks and tests of directional forecasts

Makridakis et al. (2018) show that complex forecasting methods often do not beat simple statistical benchmarks, and that this comparison is often not made at all. Pesaran and Timmermann (1992) test whether predicted and realized directions are independent. This is the hypothesis that a directional forecast really has to reject, because a hit rate above one half is not enough when up days and down days are not equally common. Diebold and Mariano (1995) compare two forecasts through their loss differences on the same observations, and Newey and West (1987) give the autocorrelation-consistent variance that we use for these differences. For pooled statistics, Sharpe ratios and costs we use the stationary bootstrap of Politis and Romano (1994), and we resample dates rather than asset-days. For per-asset hit rates we use the score interval of Wilson (1927).

2.3. Short-horizon predictability in cryptocurrency returns

Short-horizon reversal is well documented in equities (Jegadeesh, 1990; Lo and MacKinlay, 1990). In cryptocurrency markets, Caporale and Plastun (2019) find unusually large price moves on the day after an overreaction, but no strategy that profits from them, and Liu and Tsyvinski (2021) find time-series momentum at weekly horizons. Machine-learning studies of cryptocurrency direction report results that depend on the horizon, the inputs and the specification. Jaquart et al. (2021), in this journal, compare several model families at horizons of one to sixty minutes. A long–short strategy based on their forecasts earns high returns before transaction costs and negative

returns after them. Gradojevic et al. (2023) use technical indicators at hourly and daily frequencies and find that forecasting performance depends on the horizon and the specification. For equities, Gu et al. (2020) find large gains from trees and neural networks, which they link to nonlinear interactions among many predictors. With a small single-asset feature set we find the opposite ranking: logistic regression is more accurate than boosted trees on all three assets. Fama (1970) and Timmermann and Granger (2004) give the prior that we start from. Predictability may exist, but it should be small, unstable and limited by trading costs.

2.4. Leakage

Kaufman et al. (2012) define leakage as the use of information that would not be available at prediction time. They stress that a variable must be judged against the timing of the prediction task and not by its name. Bergmeir and Benítez (2012) and Cerqueira et al. (2020) study when cross-validation is valid for time series. The provenance screen in Section 3.3 deals with a case that is inside these definitions but outside the usual checks, namely a feature whose name suggests an external source but whose values are computed from the prices themselves.

3. The proposed framework

3.1. Overview

Figure 1 gives a summary of the framework and Algorithm 1 lists its steps. Two design principles are used in every layer. The first is strict information timing. Nothing that becomes known only after the close of day t may enter a forecast made at that close, and every forecast is stored with its decision date, label date and fold. The second is independent verification. Every important quantity is either computed twice, along different code paths, or checked against a fixture with a known answer. Section 4 gives the statistical and economic details. This section describes the parts of the framework that protect the computation itself.

3.2. Implementation contracts

Errors in backtest code are easy to miss, because a faulty engine still produces numbers that look reasonable. For this reason we write the rules of the engine as ten contracts in four groups, together with two reporting conventions (Table 1). Each contract protects against one specific pitfall. Several of these pitfalls came up while we were building the system, and this is why we test for them directly. Every contract is enforced by at least one regression test with a known answer. For eight of them we go one step further

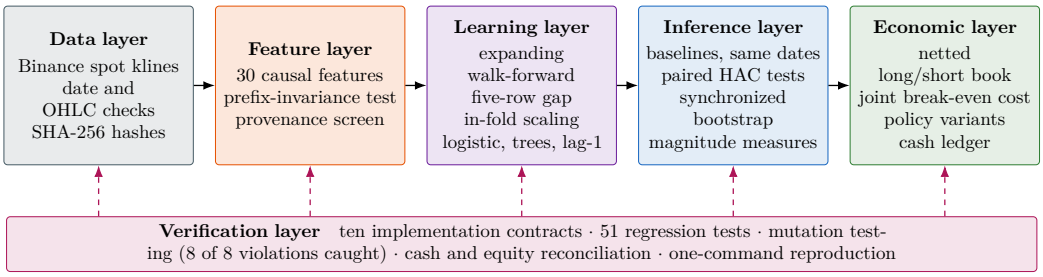


Figure 1: The evaluation framework. Forecasts move from left to right; the verification layer checks the computations in every layer.

Algorithm 1 Verified walk-forward evaluation

Require: daily bars for assets $j \in \mathcal{J}$; initial window $m = 1{,}000$; test block $b = 250$; gap $e = 5$; cost c; bootstrap replicates B and mean block length L

Validate dates, prices and OHLC bounds; record the SHA-256 hash of each input

Build causal features $x_{j,t}$ and labels $y_{j,t} = \mathbf{1}(C_{j,t+1} > C_{j,t})$; drop warm-up rows

Screen each external feature; exclude it if its rank correlation with same-day returns exceeds τ

for each asset j and fold $k = 0, 1, \ldots$ with $a_k = m + e + bk < n_j$ **do**

    Fit scaler and classifier on rows $i < a_k - e$; save the fitted pipeline

    Forecast rows $a_k \le i < \min(a_k + b,\, n_j)$; record all baseline forecasts on the same dates

end for

Per asset: Wilson intervals, Pesaran–Timmermann tests, paired HAC differences against inverse persistence

Pooled: date-synchronized stationary bootstrap with B replicates and mean block length L

Magnitude: return-weighted hit rates and accuracy by size of the next-day move

Portfolio: $w_{j,t} = s_{j,t}/J_t$; gross return g, turnover q, net return $g - cq$; Sharpe ratios and $c^* = \bar{g}/\bar{q}$ with joint bootstrap intervals

Cash ledger: book every leg with its own fee; assert agreement with the closed-form return

Verification: run the regression tests; re-insert each contract violation and require the named test to fail

Ensure: dated forecasts, fitted pipelines, inference tables, cost curve and verification report

and confirm, by mutation testing, that putting the violation back makes the named test fail (Section 6).

Table 1: Implementation contracts of the backtest engine. *Mutation*: confirmed by re-inserting the violation (Table 10). *Test*: enforced by a dedicated regression test. *Definition*: a reporting convention.

ID	Contract	Pitfall it prevents	Check
Timing			
C1	Prices are looked up at or before the signal time	Fills against a bar that has not closed	Mutation
C2	Orders fill at the open of the next bar	Same-bar execution	Test
C3	Stops and targets are checked against each bar's high and low	Missed intrabar exits	Mutation
Accounting			
C4	Each position is valued against its own asset's prices	Cross-asset pricing	Test
C5	Short positions lose value when their price rises	Inverted short P&L	Mutation
C6	Fees are charged on each leg's own notional	Cash that disagrees with reported fees	Mutation
Statistics			
C7	Sharpe ratios use equity-curve returns at the bar frequency	Annualizing per-trade returns	Mutation
C8	Monte Carlo resamples trades with replacement	A simulation whose paths are all identical	Mutation
Estimation sample			
C9	The scaler is fitted on training rows only and saved with the model	In-sample or single-row scaling	Mutation
C10	Training and test samples are separated by a gap	Overlap between training and test data	Mutation
Reporting conventions			
R1	Calmar ratio uses the maximum peak-to-trough drawdown	An unrelated denominator	Definition
R2	Accuracy-derived quantities are not labeled as risk or profit measures	Misleading metric names	Definition

Timing. Prices are looked up only at or before the signal time (C1). This is done by a binary search over $\{i : t_i \le t_{\text{signal}}\}$ and not by a nearest-bar search, which can pick a bar that has not closed yet. Orders fill at the open of the next bar (C2). Stops and targets are checked against the high and low of every bar (C3). When one bar touches both, we assume that the stop fills first, because daily OHLC data cannot show the order.

Accounting. Each position is valued with the prices of its own asset (C4), and the portfolio value is

$$V_t = \text{cash}_t + \sum_{j \in \text{long}} q_j \, p_t^{(j)} - \sum_{j \in \text{short}} q_j \, p_t^{(j)}, \tag{1}$$

so a short position loses value when its price goes up (C5). Fees are charged on the own notional of each leg (C6), so the cash that is charged always equals the reported fees.

Statistics. Sharpe ratios are computed from equity-curve returns that are sampled at the bar frequency (C7),

$$S = \frac{\overline{r - r_f}}{\text{sd}(r - r_f)} \sqrt{P}, \qquad P = \frac{\text{seconds per year}}{\text{median}\{t_{i+1} - t_i\}}, \tag{2}$$

and not by annualizing per-trade returns with a fixed factor such as $\sqrt{252}$, which is undefined when trades do not arrive on a fixed clock. All research results in this paper set $r_f = 0$. Monte Carlo analysis resamples trades with replacement and tracks drawdown (C8). Simply permuting and summing trade profits would give the same total on every path.

Estimation sample. The feature scaler is fitted on training rows only and is saved with the model (C9). If it were fitted at inference time on a single row, it would map every feature to zero. Training and test samples are separated by a gap (C10),

$$\mathcal{T}_{\text{train}} = [0, k), \qquad \mathcal{T}_{\text{test}} = [k + e, n), \qquad e \ge L_{\text{seq}} + h, \tag{3}$$

where L_{seq} is the length of any sequence window and h is the label horizon (López de Prado, 2018).

Reporting conventions. The Calmar ratio divides the annualized return by the maximum peak-to-trough drawdown (R1). Quantities derived from accuracy are never called risk or profit measures (R2). For accuracy a, a "Sharpe ratio" built by scoring correct calls as $+1$ and wrong calls as -1 equals $(2a - 1)/\left(2\sqrt{a(1-a)}\right)$, so it carries no information beyond a. These two conventions are definitions and not computations, so we state them without separate tests.

Independent checks. When two metrics are produced along different code paths, errors in backtest code can hide each other. For example, a sign error in the valuation of short positions corrupts the equity curve, but it never reaches a Sharpe ratio that is computed from the trade list. So the framework reconciles each economic quantity with an independent calculation: the equity curve with the summed trade profits, the cash ledger with a closed-form return formula, and the saved models with the stored forecasts.

3.3. Provenance screening

Some features come with names that promise information from outside the market, such as news sentiment. Before such a feature can enter a model, the framework checks how well it can be rebuilt from prices of the same day. We show the screen on a synthetic sentiment series that was generated from daily prices as a placeholder during development, because free news archives usually go back only about 30 days. For each daily bar, the generator computes the open-to-close return $u_t = (C_t - O_t)/O_t$ and sets

$$s_t = \mathrm{round}_3\Big[\mathrm{clip}\Big(\mathrm{clip}(15u_t, -1, 1) + \epsilon_t,\ -1,\ 1\Big)\Big], \qquad \epsilon_t \sim \mathcal{U}(-0.2, 0.2), \quad (4)$$

and then adds a label (positive or negative beyond ± 0.3, neutral otherwise) and a template headline.

Table 2: Provenance screen for the synthetic sentiment series: dependence on the same day's open-to-close return and on the next day's close-to-close return. Sign agreement is the share of days on which the two have the same sign.

		Same day, u_t			Next day, r_{t+1}	
Asset	Days	Pearson	Rank	Signs agree	Pearson	Signs agree
BTC	3,150	0.916	0.945	87.0%	−0.018	47.2%
ETH	3,150	0.907	0.968	89.2%	−0.034	46.2%
DOGE	2,463	0.519	0.972	90.1%	−0.015	46.3%

Table 2 reports the screen. The score has the same sign as the same day's return on 87–90% of days and has rank correlations of 0.95–0.97 with it, while it has almost no relation with the next day's return. So the screen flags it on every asset, and the framework keeps it out of all models. The choice of statistic is important here. A linear fit on the same-day return explains 84% of the score's variance for BTC and 82% for ETH, but only 27% for DOGE, because the large daily moves of DOGE often push $15u_t$ past the clipping bounds. A linear threshold would therefore have passed the DOGE series, while its rank correlation of 0.97 flags it. If such a score is used to explain

the same day's direction, it is the same as using the target as a feature. If it is used after the close to forecast the next day, it does not leak future information, but it adds nothing that prices do not already contain.

4. Data and experimental design

4.1. Instrument and data

We use daily Binance spot klines for BTCUSDT, ETHUSDT and DOGEUSDT, quoted in USDT (Binance, 2026). Each bar covers one UTC calendar day and is fully known only after 24:00 UTC. The fields are the open, high, low and close prices, base and quote volume, number of trades and taker-buy base volume. It is important to name the instrument, because fees, shorting and funding are different for spot, margin and perpetual futures. The long/short strategies take short positions, which need margin on the spot market. We treat this as an idealization and come back to it in Section 8.

The BTC and ETH files run from 17 August 2017 to 1 April 2026 (3,150 days each), and the DOGE file runs from 5 July 2019 to 1 April 2026 (2,463 days). No date is missing or repeated, all prices are positive, and every bar satisfies $L_t \leq \min(O_t, C_t) \leq \max(O_t, C_t) \leq H_t$. As a provenance check, we compared all eight numeric fields for 275 asset-days (July 2019, January 2023 and March 2026 for each asset) with the checksum-verified monthly archives of the exchange, and every field matched to a relative tolerance of 10^{-10}. This check covers the sampled months, not the whole history.

4.2. Sample construction

Table 3: Sample reconciliation. Usable rows exclude the 60-day feature warm-up and the last two days of each file. Out-of-sample (OOS) forecasts start after the first 1,005 usable rows (1,000 for initial training plus a five-row gap). The panel has 5,562 asset-day forecasts on 2,083 distinct decision dates.

Asset	Data period	Raw rows	Usable rows	OOS forecasts	Folds	OOS decision dates
BTCUSDT	2017-08-17 to 2026-04-01	3,150	3,088	2,083	9	2020-07-17 to 2026-03-30
ETHUSDT	2017-08-17 to 2026-04-01	3,150	3,088	2,083	9	2020-07-17 to 2026-03-30
DOGEUSDT	2019-07-05 to 2026-04-01	2,463	2,401	1,396	6	2022-06-04 to 2026-03-30
Panel				**5,562**	24	2,083 dates

Three sample definitions are used in this paper (Table 3). Raw rows are the rows of the input files. Usable rows drop the first 60 days, which the longest rolling windows need, and the last two days. The final decision date is 30 March 2026, whose label uses the close of 31 March, and 1 April is an unused buffer that we keep so that all three assets share one cutoff. Usable rows are therefore raw rows minus 62. Out-of-sample forecasts begin after

the first 1,005 usable rows. The panel has 5,562 asset-day forecasts but only 2,083 distinct decision dates: two assets on the 687 dates before DOGE enters (17 July 2020 to 3 June 2022) and three assets on the other 1,396 dates. BTC and ETH move together, so asset-days are not independent observations, and the inference in Section 4.6 resamples dates and not asset-days.

4.3. Features

The models use 30 features, and each one is computed from bars dated t or earlier (Appendix A). They are log returns over 1, 2, 3, 5, 10 and 20 days; rolling volatility and close-price z-scores over 5, 10, 20 and 60 days; three measures of the shape of the bar; seven trading-activity measures built from the kline fields (taker-buy share, average trade size, trade count and volume, as z-scores or ratios); and RSI, a MACD histogram, Bollinger-band position, normalized ATR and a day-of-week encoding. A test checks the number of features, and it also checks that the features for date t do not change when later rows are deleted. This is a direct check of causality.

This is a single-asset technical feature set. It does not include the inputs that are most specific to cryptocurrency markets, such as perpetual funding rates, open interest, basis, liquidations, order-book imbalance, exchange flows and lead–lag effects across assets. This makes it a conservative test of what price and activity data alone can give.

4.4. Target, validation and models

The target is $y_t = \mathbf{1}(C_{t+1} > C_t)$. Ties happen on 6 of the 5,562 evaluation asset-days, and they are put in class zero. Validation is an expanding-window walk-forward (Figure 2). With usable rows indexed $i = 0, \ldots, n-1$, fold k tests rows $a_k \le i < \min(a_k + 250, n)$, where $a_k = 1{,}005 + 250k$, and trains on rows $i < a_k - 5$. A final block is kept only if it has at least 30 rows. BTC and ETH have nine folds (eight of 250 days and a last one of 83), and DOGE has six (five of 250 and one of 146). The five excluded rows separate the last training label from the first test forecast. For example, in the first BTC fold the last training label is the close of 12 July 2020, and the first test forecast is made at the close of 17 July. Rolling features in a test fold do share past bars with the training data, but shared history is not future information. Every forecast is made only once, out of sample.

Within each fold, a standard scaler and a classifier are fitted together as one pipeline on the training rows, and the pipeline is saved. The scaler is never refitted at prediction time. The main model is an ℓ_2-regularized logistic regression (inverse penalty $C = 0.1$, with intercept, at most 2,000 iterations). The second model is a histogram gradient-boosted tree classifier (depth 3, 200 iterations, learning rate 0.05, ℓ_2 leaf regularization 1, no early

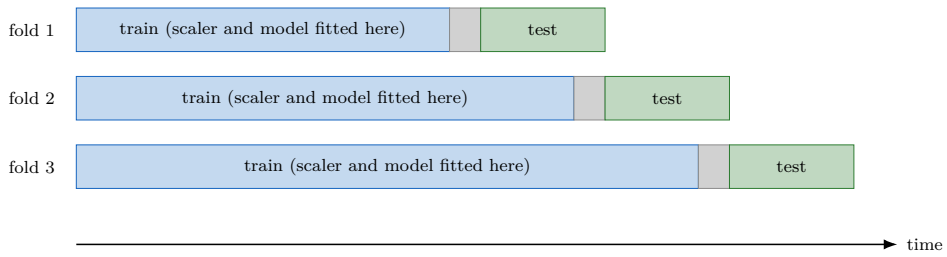


Figure 2: Expanding-window walk-forward evaluation. The training window grows by 250 rows per fold, and the scaler and classifier are refitted inside each fold on its training rows only.

stopping, seed 0). A logistic regression that sees only $\log(C_t/C_{t-1})$ is used as an ablation. A forecast is "up" when the predicted probability is above 0.5. The hyperparameters were fixed in advance and kept the same across assets and folds, and none of them was tuned on test data. The study is retrospective and was not preregistered. In total, the evaluation fits 72 pipelines (24 asset-folds times three classifiers).

4.5. Baselines

All baselines are scored on exactly the same dates as the model.

- **Persistence** predicts that the next move repeats the latest completed move, $\hat{y}_t = \mathbf{1}(C_t > C_{t-1})$.
- **Inverse persistence** predicts the opposite, $\hat{y}_t = 1 - \mathbf{1}(C_t > C_{t-1})$, so exactly one of these two rules is right on any day. It captures one-day reversal, and in these data it turns out to be the strongest simple benchmark.
- **Training majority** predicts the more frequent class in the current training fold. In practice it was long BTC and short DOGE all the time, and long ETH on 88% of days.
- **Passive basket** is an equal-weight long position in the assets that are available on each date, rebalanced daily. It is a return benchmark, not a classifier.

4.6. Statistical inference

For per-asset hit rates we report Wilson intervals, a two-sided test against 0.5, and the Pesaran–Timmermann test of independence between predicted and realized direction.

The comparison with inverse persistence is paired. For a model M and inverse persistence I, define

$$d_{j,t} = \mathbf{1}\big(\hat{y}^{M}_{j,t} = y_{j,t}\big) - \mathbf{1}\big(\hat{y}^{I}_{j,t} = y_{j,t}\big). \tag{5}$$

For each asset we estimate the mean of d with a Bartlett-kernel HAC variance,

$$\widehat{\mathrm{Var}}(\bar{d}) = \frac{1}{n}\Big[\hat{\gamma}_0 + 2\sum_{\ell=1}^{K}\Big(1 - \frac{\ell}{K+1}\Big)\hat{\gamma}_\ell\Big], \qquad \hat{\gamma}_\ell = \frac{1}{n}\sum_{t=\ell+1}^{n} v_t v_{t-\ell}, \qquad K = 20, \tag{6}$$

where $v_t = d_t - \bar{d}$, and we report normal-approximation intervals and two-sided p-values.

For pooled statistics, Sharpe ratios and costs we use a circular stationary bootstrap with 2,000 replicates, a mean block length of 20 days and seed 20,260,401. Block lengths of 5 and 60 days are used as sensitivity checks. The bootstrap resamples dates, not asset-days. In each replicate one sequence of date indices is drawn and applied to every asset, including the mask that records which assets are active, so the dependence between assets on the same day is kept. Pooled accuracy and the pooled paired difference are computed again in each replicate as ratios of sums over the sampled dates. Forecasts are not refitted inside the replicates, so the intervals are conditional on the fitted walk-forward predictions. We report percentile intervals, and we treat secondary comparisons as exploratory.

4.7. Economic evaluation

Target-notional book. On date t, let A_t be the set of assets with a forecast and $J_t = |A_t|$. Each active asset gets the target weight $w_{j,t} = s_{j,t}/J_t$, with $s_{j,t} = 2\hat{y}_{j,t} - 1 \in \{-1, +1\}$, and inactive assets get zero. Then

$$g_{t+1} = \sum_j w_{j,t}\, r_{j,t+1}, \qquad q_t = \sum_j \big|w_{j,t} - w_{j,t-1}\big|, \qquad n_{t+1}(c) = g_{t+1} - c\, q_t, \tag{7}$$

where $r_{j,t+1} = C_{t+1}/C_t - 1$. Weights start at zero, and the turnover of the last day includes closing the book. Reversing the position in one asset costs two units of turnover, and positions that keep their sign are carried over without trading. The reference cost is $c = 9$ basis points (bps) per unit of one-way turnover. It is made up of a 4-bps taker fee and 5 bps of slippage, and Section 5.6 reports the whole cost curve. The book is directional and does not need to be market neutral.

Sharpe ratios are annualized with 365 periods, the sample standard deviation and a zero risk-free rate, and we also report annualized arithmetic

means and volatilities. In this linear book, the cost at which the mean net return is zero is

$$c^* = 10^4\,\bar{g}/\bar{q} \text{ bps}, \qquad \bar{q} > 0. \tag{8}$$

Each bootstrap replicate resamples g and q together and computes c^* and both Sharpe ratios again, so the uncertainty about turnover is included in the interval. Values below zero are kept and not clipped.

Cash ledger. The second calculation is an explicit cash account with an execution rule that is conservative on purpose. A forecast made at the close of t is entered at the open of $t+1$ and closed at the close of $t+1$. This happens every day, even when the next forecast points in the same direction, so the ledger pays a full round trip on each asset every day. Each asset gets $1/J_t$ of the equity at the start of the day. For side $s \in \{-1, +1\}$, slippage $\lambda = 5$ bps and fee rate $f = 4$ bps,

$$\begin{aligned} P_{\text{in}} &= O_{t+1}(1 + s\lambda), \qquad P_{\text{out}} = C_{t+1}(1 - s\lambda), \\ R &= s\Big(\frac{P_{\text{out}}}{P_{\text{in}}} - 1\Big) - f\Big(1 + \frac{P_{\text{out}}}{P_{\text{in}}}\Big). \end{aligned} \tag{9}$$

The ledger books the entry and exit cash flows and the fee of each leg separately, and it checks them against (9). Positions are flat at every close, the account starts with 10,000 units, and the Sharpe ratio uses only the 2,083 evaluation days. Figure 3 shows how the return windows line up.

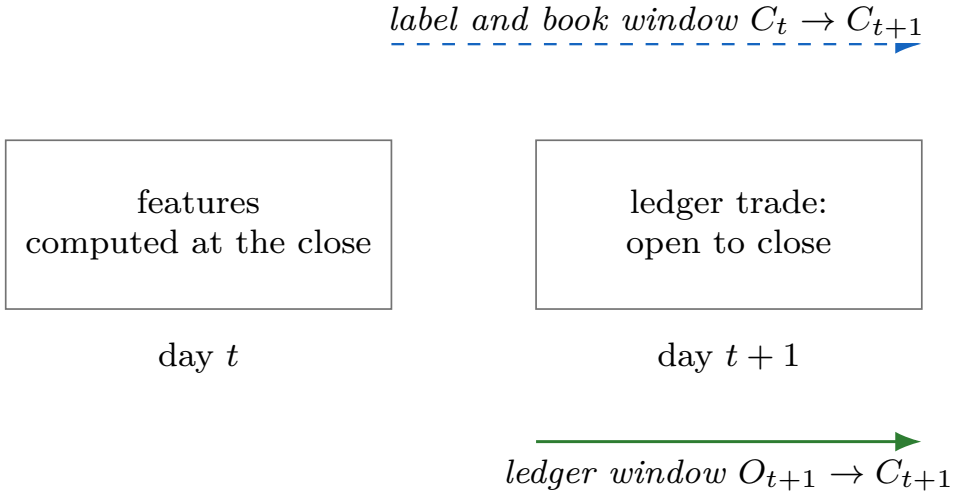


Figure 3: Return windows. Because each day opens within a few basis points of the previous close, $O_{t+1} \approx C_t$ and the ledger window closely tracks the label window.

4.8. Reproducibility

Every number in this paper comes from deterministic scripts. One command rebuilds the 72 pipelines, writes the 5,562 dated forecasts with their fold identifiers, probabilities and baseline decisions, computes all tables and the supplementary diagnostics, reconciles the trade ledger, and runs the regression and mutation suites (Appendix E).

5. Results

5.1. Directional accuracy

Table 4: Out-of-sample directional accuracy (%). Pooled values are shares of correct asset-days.

Predictor	BTC	ETH	DOGE	Pooled
Logistic regression	**52.71**	**53.10**	**53.08**	**52.95**
Boosted trees	52.09	51.37	51.93	51.78
Lag-1 logistic (ablation)	51.90	52.66	52.58	52.36
Inverse persistence	53.00	52.57	53.15	52.88
Persistence	47.00	47.43	46.85	47.12
Training majority	50.55	49.59	51.72	50.49
Forecasts (n)	2,083	2,083	1,396	5,562

Table 4 reports the out-of-sample accuracy of every predictor. Logistic regression is the most accurate of the learned models. It is correct on 52.95% of the 5,562 asset-days, and the date-synchronized bootstrap puts the 95% interval for this pooled number at 51.45–54.41%. This is clearly above one half, even after we allow for the dependence between assets. The result also holds for each asset. The Wilson intervals are $[50.6, 54.8]$, $[50.9, 55.2]$ and $[50.5, 55.7]$ for BTC, ETH and DOGE, and the Pesaran–Timmermann test rejects independence of predicted and realized direction with p-values of 0.015, 0.008 and 0.016. Logistic regression is more accurate than gradient-boosted trees on all three assets. This suggests that, with a modest single-asset feature set, the extra flexibility of trees does not help much.

5.2. Comparison with the reversal benchmark

Next, the framework asks whether the model knows more than a rule with no parameters. Inverse persistence is correct on 52.88% of asset-days, which is almost the same as the model, and Table 5 compares the two on paired observations. The logistic model is ahead on ETH by 0.53 points and within 0.3 points on BTC and DOGE. The pooled difference is +0.07 points,

Table 5: Paired accuracy differences against inverse persistence on identical dates, in percentage points. A positive difference favors the model. Per-asset intervals use a 20-lag Bartlett HAC variance; the pooled interval uses the date-synchronized stationary bootstrap (mean block length 20).

Asset	Model	Difference	95% interval	p
BTC	Logistic	−0.29	[−2.55, 1.97]	0.803
BTC	Boosted trees	−0.91	[−3.03, 1.21]	0.399
BTC	Lag-1 logistic	−1.10	[−3.31, 1.10]	0.327
ETH	Logistic	+0.53	[−1.85, 2.91]	0.663
ETH	Boosted trees	−1.20	[−3.73, 1.33]	0.353
ETH	Lag-1 logistic	+0.10	[−1.87, 2.06]	0.924
DOGE	Logistic	−0.07	[−2.69, 2.55]	0.957
DOGE	Boosted trees	−1.22	[−3.84, 1.41]	0.363
DOGE	Lag-1 logistic	−0.57	[−1.91, 0.76]	0.399
Pooled	Logistic	+0.07	[−1.44, 1.61]	–

with a 95% interval from −1.44 to 1.61, and the interval hardly changes with five-day or sixty-day blocks (Table 8). So in terms of accuracy, the model and the reversal rule cannot be separated. We see this as a finding about the data and not as a weakness of the model. In daily cryptocurrency returns, much of the predictable direction is one-day reversal, and a well-regularized model finds it without being told to. The next two subsections show what the model learns beyond that rule and where its forecasts add value.

5.3. What the model learns

The logistic model agrees with inverse persistence on 70.8% of asset-days (69.1% for BTC, 71.1% for ETH and 73.0% for DOGE). Table 6 shows the reason. Three of the features describe the move of the same day: the one-day log return, the open-to-close return and the position of the close within the day's range. For BTC and ETH all three have negative coefficients in every fold. The close location has the largest weight of all 30 features for BTC (−0.17) and the second largest for ETH (−0.13). These features overlap a lot. The one-day and open-to-close returns have correlations of 0.998, 0.997 and 0.915 for BTC, ETH and DOGE, because each day opens within a few basis points of the previous close, and the close location is correlated with both (about 0.7 for BTC and ETH). So their coefficients are best read together, as a regularized description of one reversal signal. For DOGE the same signal shows up as the largest weight of any feature, a negative one on the one-day return (−0.42), which is partly offset by a positive weight on the open-to-close

return (+0.26).

Table 6: Selected standardized logistic coefficients, averaged over folds. In parentheses: the number of folds in which the coefficient has the same sign as its average. Features are standardized within each training fold, so magnitudes are comparable within a column.

Feature	BTC (9 folds)	ETH (9 folds)	DOGE (6 folds)
One-day log return	−0.045 (9)	−0.100 (9)	−0.415 (6)
Open-to-close return	−0.042 (9)	−0.073 (9)	+0.259 (6)
Close location in day's range	−0.174 (9)	−0.132 (9)	+0.011 (4)
Three-day log return	+0.035 (9)	+0.056 (8)	+0.248 (6)
Close z-score (60 days)	+0.126 (9)	+0.039 (9)	+0.023 (5)
Average trade size	−0.131 (9)	+0.001 (4)	−0.053 (6)
Normalized ATR (14 days)	+0.038 (8)	+0.187 (9)	−0.053 (6)
Trade-count z-score (60 days)	−0.058 (7)	+0.130 (9)	−0.079 (5)
Weekday, $\cos(2\pi d/7)$	−0.057 (9)	−0.075 (9)	−0.128 (6)

Beyond reversal, the model also uses information that the reversal rule cannot see. The three-day return has a positive sign for every asset (in 23 of the 24 asset-folds), which is a mild continuation over several days. For BTC, the second and third largest weights go to average trade size (negative) and to the close relative to its 60-day mean (positive), each with the same sign in every fold. For ETH, the largest weight of all goes to normalized ATR, and the trade-count z-score ranks third. A weekly seasonal term is negative for every asset in every fold. These terms are what make the forecasts of the model different from the reversal rule on the other 29% of asset-days.

5.4. Where the model adds value: magnitude

Directional accuracy gives a 0.2% day the same weight as a 12% day, so it is a poor guide to profit in a market that is this volatile. Table 7 looks at the size of the moves.

Panel A splits the evaluation days of each asset into terciles of the absolute next-day return. The reversal rule does better on small and medium days, but on the largest third of days the model is correct 51.1% of the time, against 48.9% for the rule. Panel B shows what this means for returns. The model has a higher return-weighted hit rate than the reversal rule on all three assets, and it is the only rule in the panel whose single-asset gross Sharpe ratio is positive on every asset: 0.44 for BTC, 0.42 for ETH and 0.61 for DOGE. The reversal rule, on the other hand, loses money on DOGE. There its return-weighted hit rate is 47.0% and its gross Sharpe ratio is −0.79, even with a 53.2% hit rate, so on DOGE accuracy and returns rank the two rules in opposite orders. The lag-1 ablation, which uses only reversal, shows the same pattern. This

Table 7: Accuracy and magnitude. Panel A: accuracy (%) by within-asset tercile of the absolute next-day return, all assets pooled. Panel B: return-weighted hit rate (%), $\sum_t |r_t| \mathbf{1}(\text{correct}_t) / \sum_t |r_t|$, and the gross annualized Sharpe ratio of a single-asset position that is long when the forecast is up and short otherwise.

Panel A. Accuracy by size of the next-day move

	Small	Medium	Large
Asset-days	1,856	1,853	1,853
Logistic regression	52.9	54.9	**51.1**
Inverse persistence	54.5	55.3	48.9

Panel B. Magnitude-aware measures by asset

	Return-weighted hit rate			Gross Sharpe ratio		
Predictor	BTC	ETH	DOGE	BTC	ETH	DOGE
Logistic regression	**51.7**	**51.6**	**52.3**	+0.44	+0.42	+**0.61**
Lag-1 logistic	52.6	52.6	47.1	+0.67	+0.69	−0.77
Inverse persistence	51.1	50.9	47.0	+0.30	+0.25	−0.79
Persistence	48.9	49.1	53.0	−0.30	−0.25	+0.79

suggests that the additional terms of the model (Section 5.3) are what keep it on the right side of the large moves of DOGE.

The same pattern holds in the portfolio. The logistic book has a gross Sharpe ratio of 0.54 and the inverse-persistence book has 0.004 (Table 9). When dates are resampled jointly, the annualized gross mean of the logistic book is higher than that of the reversal book by 31.5 percentage points. The 95% interval, from −15.0 to +76.9, is wide, so the size of the advantage is uncertain, but its direction is the same in all the magnitude measures.

5.5. Stability across folds

Accuracy changes with market conditions (Appendix B). For the logistic model it reaches 61.6% in the fifth fold of ETH (April to December 2023) and 59.2% in the sixth fold of BTC (December 2023 to August 2024), and it ranges from 42.2% to 61.6% across the 24 asset-folds. The most recent folds, which are also the shortest (83 days for BTC and ETH and 146 for DOGE), are the weakest for both the model and the reversal rule. So a pooled number is an average over quite different regimes, and we publish the fold-level record together with the results.

5.6. Economic performance

Table 8 reports the target-notional book for the logistic forecasts. Before costs it earns an annualized mean of 31.7% with a volatility of 58.4%, which is

Table 8: Target-notional book for the logistic forecasts. Panel A: point estimates at 9 bps per unit of one-way turnover over 2,083 dates. Panel B: 95% percentile intervals from the date-synchronized stationary bootstrap (2,000 replicates) for three mean block lengths; 20 days is the primary choice. The accuracy difference is against inverse persistence, in percentage points.

Panel A. Point estimates

	Annualized mean	Volatility	Sharpe ratio
Gross	31.7%	58.4%	0.544
Net	6.6%	58.4%	0.113

Turnover 0.766 a day; cost 6.89 bps a day; break-even cost $c^* = 11.36$ bps.

Panel B. 95% intervals by mean block length

Block	Accuracy diff.	Gross Sharpe	Net Sharpe	Break-even (bps)
5 days	[−1.49, 1.62]	[−0.27, 1.30]	[−0.69, 0.86]	[−5.68, 26.95]
20 days	[−1.44, 1.61]	[−0.19, 1.32]	[−0.60, 0.87]	[−4.00, 26.43]
60 days	[−1.47, 1.56]	[−0.14, 1.30]	[−0.56, 0.84]	[−3.06, 25.78]

a Sharpe ratio of 0.544. Turnover is 0.766 units a day on average, so at 9 bps the cost is 6.89 bps a day. This leaves a net mean of 6.6% and a net Sharpe ratio of 0.113. The break-even cost is 11.4 bps per unit of turnover, which is above the 9 bps that we assume, so the strategy stays profitable in the point estimate at realistic fee and slippage levels (Figure 4). The bootstrap intervals are wide: −0.19 to 1.32 for the gross Sharpe ratio, −0.60 to 0.87 for the net Sharpe ratio and −4.0 to 26.4 bps for the break-even cost. They change little with the block length. This is what we should expect from 2,083 daily observations, for which a Sharpe ratio of 0.54 has a standard error of roughly 0.38, and it is the reason why the framework gives an interval for every economic quantity.

5.7. Policy variants

Table 9 applies the same book to every predictor and to two variants of the logistic policy. Among the predictors, logistic regression has both the highest gross and the highest net Sharpe ratio. Its closest rival on a net basis is the training-majority rule. This rule is in effect a static allocation (long BTC, short DOGE and mostly long ETH) with almost no turnover. Boosted trees, the lag-1 ablation and both persistence rules all earn less before costs or trade more, and the persistence rules, which reverse their position on 53% of days, are hit hardest by costs. Two variants improve the logistic book further. If we restrict it to the 1,396 dates on which all three assets are active,

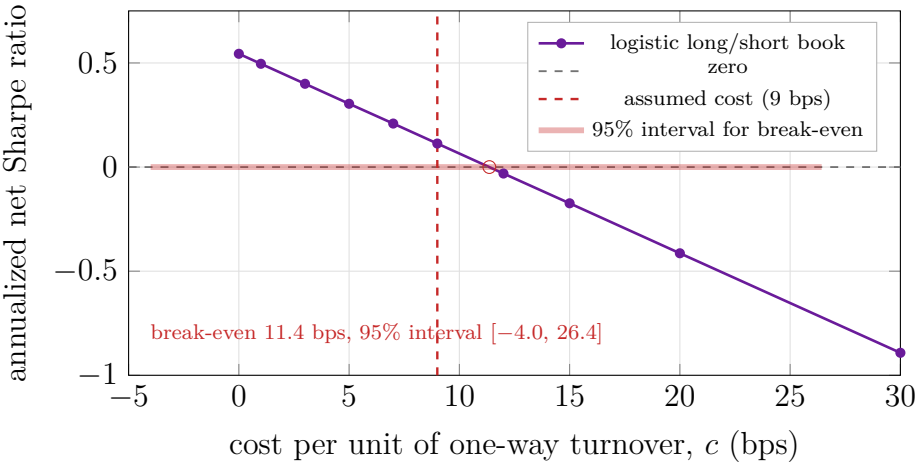


Figure 4: Net Sharpe ratio of the logistic book as a function of the cost per unit of turnover. The curve is a set of point estimates, not a confidence band; the shaded segment is the bootstrap interval for the break-even cost.

Table 9: Target-notional book for every predictor and two variants of the logistic policy, at 9 bps per unit of turnover. Intervals were computed for the primary logistic book (Table 8); the remaining rows are descriptive.

Policy	Dates	Gross Sharpe	Net Sharpe	Turnover
Logistic regression	2,083	**+0.544**	**+0.113**	0.766
Boosted trees	2,083	+0.234	−0.216	0.770
Lag-1 logistic	2,083	+0.386	−0.057	0.817
Inverse persistence	2,083	+0.004	−0.547	1.060
Persistence	2,083	−0.004	−0.555	1.060
Training majority	2,083	+0.112	+0.110	0.002
Logistic, three-asset dates	1,396	+0.947	+0.473	0.755
Logistic, long or cash	2,083	+0.964	+0.712	0.383

the net Sharpe ratio rises to 0.47. A long-or-cash version that never sells short reaches 0.71 with half the turnover, and it holds a position on 56% of asset-days. For context, the fully invested passive basket had a gross Sharpe ratio of 0.92 over this strongly rising period. Confidence gating, which trades only when the predicted probability is far from one half, lowers turnover, but it did not raise the net Sharpe ratio in this sample (Appendix D).

5.8. Execution policy

The cash ledger gives a second view of implementation, which is conservative on purpose. Before costs, the logistic forecasts point the right way

on 52.9% of next-day open-to-close moves and capture 9.9 bps per trade on average. So the forecasting edge carries over from the close-to-close label window to an open-to-close window that can be traded. However, the ledger closes and reopens every position every day, so it pays about 18 bps per asset-day in slippage and fees, against 6.9 bps a day for the netted book. At this rate the edge is used up: the net return per trade is -8.1 bps and the Sharpe ratio of the ledger is -0.58. This comparison is useful. Entry timing costs almost nothing, because each day opens within a few basis points of the previous close. Over the evaluation window the standard deviation of the close-to-next-open return is 0.6, 0.9 and 3.5 bps for BTC, ETH and DOGE, against 306, 415 and 1,084 bps for the open-to-close return. What decides whether the forecasts can be traded is whether the execution rule nets consecutive positions, as the target-notional book does. The ledger is also an accounting check. Its cash reconciles with the trade records to within 5×10^{-12} units on every day, and every position is flat at the close.

5.9. Relation to passive exposure

A directional long/short book should not be ranked against a long-only holding by raw Sharpe ratios. What matters is how the two work together. When we regress the net daily return of the logistic book on the passive basket, we get a beta of 0.17 and a correlation of 0.20. So the strategy has little market exposure, and it is a natural candidate for diversification. Its regression intercept is small and not statistically different from zero ($t = -0.18$ with a 20-lag HAC standard error). Over this sample, its contribution to a passive holding is therefore mainly diversification, not a measurable stand-alone alpha.

6. Verification of the implementation

The number of tests shows effort, but what matters is whether the tests find the failures they were written for. The test suite for the backtest engine and the feature pipeline has 29 tests. They cover next-bar entry, asset-specific pricing, short valuation, intrabar stops, the holding rule, annualization, resampling, cash reconciliation, scaler persistence and split gaps. The research code adds 22 more checks, for example a hand-computed HAC variance, rejection of degenerate inputs, the exact cost root, own-notional fees, the feature count and prefix invariance, and reconciliation of the saved models, forecasts and results. All 51 tests pass.

For mutation testing we put back one contract violation at a time and run the 29-test suite again. A mutant counts as detected only if the test written for that contract fails, the number of tests that ran is the same as in the

baseline, and the exit status agrees. Collection errors and skipped tests do not count as detection. Table 10 shows that all eight mutants are detected. The other two contracts, next-bar fills (C2) and asset-specific pricing (C4), cannot be violated on their own without rewriting the simulation loop, so each of them is protected by a dedicated test instead. Because we chose the eight mutants ourselves, the score is evidence about these contracts and not about every possible defect.

Table 10: Mutation testing. Each row re-inserts one contract violation into the engine and reruns the 29-test suite. "Failed" is the number of tests that fail; the named test is the one that must fail for the mutant to count as detected.

Contract	Re-inserted violation	Test that must fail	Failed
C1	Nearest bar, possibly in the future	`price_lookup_never_selects_a_future_bar`	1
C3	Stops checked at the close only	`stop_wins_when_one_bar_touches_both_levels`	2
C5	Short valued with the wrong sign	`short_equity_falls_when_price_rises`	1
C6	Fee split evenly across legs	`equity_curve_reconciles_with_summed_trade_pnl`	1
C7	Per-trade annualization	`annualisation_matches_the_closed_form`	2
C8	Permutation used as resampling	`bootstrap_is_not_degenerate`	1
C9	Scaler refitted at inference	`single_row_transform_is_not_all_zeros`	2
C10	Gap between folds removed	`gap_between_folds_is_at_least_the_sequence_length`	2
Mutation score			**8/8**

The pipeline can also be reproduced on different platforms. A clean re-run in a new Windows environment with Python 3.12 reproduced the primary output files byte for byte. A re-run on Linux with Python 3.13 and newer library versions reproduced every forecast and table, and the differences in predicted probabilities were below 10^{-12}. The input files are identified by SHA-256 hashes (Appendix E).

7. Discussion

7.1. What the framework reveals

Three findings stand out, and each one depends on a layer of the framework that a standard evaluation would leave out. The first comes from the baseline layer. A hit rate of 52.95% looks like a result of the model, but almost all

of it is also available to a rule with no parameters, and without the paired comparison nobody would notice this. The second comes from the magnitude layer. What the model adds over that rule is not how often it is right, but on which days it is right, and accuracy alone cannot show this difference. On DOGE, accuracy even ranks the two rules in the wrong order. The third comes from the economic layer. Whether a forecasting edge can be traded depends on costs and on the execution rule as much as on the forecast. The same forecasts earn a positive net Sharpe ratio when positions are netted and a negative one when they are rebuilt every day.

7.2. Statistical and economic significance

The logistic hit rates are different from one half on every asset, but the interval for the gross Sharpe ratio contains zero. Both statements are true. Accuracy affects returns only through a small margin, $2a - 1 \approx 0.06$, and through which days happen to be called correctly, while costs take away an amount that is proportional to turnover. With 2,083 dates, a Sharpe ratio of 0.54 has a bootstrap standard error of roughly 0.38. Finding an effect and being paid for it need different amounts of statistical power, and an evaluation should report both.

7.3. Practical guidance

Table 11 sums up the framework as a list of components, the problem that each one protects against, and the evidence for it in this study. Each component is cheap to use. Together they turn a single accuracy number into a clear account of what a model knows, where its value comes from and what it would cost to use it.

8. Limitations and future work

The framework is general, but the evidence in this paper has a clear scope. It covers three large assets that are still listed, on one exchange, with daily bars that start at 00:00 UTC. Other assets, exchanges and bar boundaries still need to be studied, and moving the start of the bars to other hours would show whether the results depend on the boundary. The analysis is retrospective. The hyperparameters were fixed in advance, but nothing was preregistered, so a prospective holdout is the natural next step. The 30 features are single-asset technical measures. Funding rates, open interest, basis, liquidations, order-book data and cross-asset effects are clear extensions. Cost-aware training objectives and meta-labeling (López de Prado, 2018) are also possible, and meta-labeling targets exactly the gap between direction and magnitude that we show in Section 5.4. The HAC estimator and the

Table 11: Framework components, what each guards against, and the evidence in this study.

Component	Guards against	Evidence here
Score persistence, inverse persistence and the majority class on identical dates	Crediting a model with an effect that a one-line rule captures	Reversal rule 52.88%, model 52.95%
Paired, dependence-aware tests; resample dates, not asset-days	Overstated significance from correlated observations	Pooled accuracy interval 51.45–54.41%
Magnitude-aware measures next to accuracy	Hit rates that hide where money is made	DOGE: reversal Sharpe -0.79, model $+0.61$
Break-even cost estimated jointly with turnover; explicit execution rule	False precision about tradability	11.4 bps, interval $[-4.0, 26.4]$; netting versus daily rebuild
Engine rules as contracts, confirmed by mutation	Tests that accompany code without checking it	8 of 8 violations detected
Rank-based provenance screen for external features	Price-derived features presented as external information	Sentiment proxy: rank correlation 0.95–0.97 with same-day return
Dated forecasts and saved pipelines	Results that cannot be re-scored	5,562 forecasts; 72 pipelines

stationary bootstrap allow for short-range dependence, but not for every kind of regime shift, and the bootstrap does not refit the models. Finally, the cost model is a flat 9 bps per unit of turnover. Maker execution could make it lower for BTC and ETH, while shorting on the spot market needs margin, and we do not model borrowing costs, funding or capacity. The ratios in this paper are research diagnostics, not expected returns for investors.

9. Conclusion

We have built a tested, leakage-safe framework for evaluating machine-learning forecasts of cryptocurrency returns, and we applied it to 5,562 daily out-of-sample forecasts for BTC, ETH and DOGE. A regularized logistic model is correct 52.95% of the time, with a dependence-robust interval that excludes one half, and it beats a coin flip on every asset. The framework shows that much of this directional skill is shared with one-day reversal, and that the special value of the model is in magnitude. It gets more of the large moves right, it has the higher return-weighted hit rate on every asset, and it

earns a gross Sharpe ratio of 0.54 against 0.004 for the reversal rule. At a realistic cost of 9 bps per unit of turnover the netted strategy keeps a positive net Sharpe ratio of 0.11, its break-even cost is 11.4 bps, and a long-or-cash version reaches 0.71. The intervals are wide, as we expect with less than six years of daily data.

The wider lesson is about evaluation. Accuracy, value beyond simple rules, value on large moves and profit after costs are different questions, and each one needs its own evidence. The framework answers all four with dated forecasts, inference that allows for dependence, cost uncertainty, and an engine whose rules are checked by tests that are themselves tested. Every number comes from one command and every fitted model is saved, so the same process can be applied directly to richer features, other assets and new models.

Data and code availability

The code, the three input files, the dated out-of-sample forecasts, the trade ledger and all machine-readable results are kept in a public code repository. To keep the review anonymous, its link is not shown in this copy. The link is given on the separate title page, and it will be added to the paper after review. The market data are public Binance kline archives. The complete analysis is rebuilt by `python scripts/reproduce_jfds.py`, and Appendix E lists the software versions, random seeds and input-file hashes.

Declaration of generative AI and AI-assisted technologies

AI use statement. During the preparation of this work the authors used OpenAI Codex and Anthropic Claude. We used these tools to help write and refactor parts of the analysis code and tests, to suggest related literature, to draft and revise parts of the manuscript, to check its wording, formatting and internal consistency, and to compare the reported numbers with the outputs of the replication scripts, which included running those scripts again. We did not use generative AI to create, change or impute data, and no number in this paper was produced by a language model. Every result comes from the deterministic scripts in the replication package, which the authors ran and checked. An author reviewed all code suggested by an AI tool before it was kept, and its behavior is checked by the regression and mutation tests in Section 6. The authors read and verified every reference suggested by an AI tool before citing it. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication, including all text, claims, code and other material produced with the help of generative AI.

Ethics statement

This study uses publicly available exchange price data and software written by the authors. It does not involve human participants, personal data or animals, so ethical approval and informed consent were not needed. The synthetic sentiment series in Section 3.3 is a development placeholder, and its generator script marks it as a price-based proxy. We use it here only to show how the provenance screen works. Nothing in this paper is investment advice. Backtested ratios like the ones in this paper are research diagnostics and not expected returns, and trading cryptocurrencies has a high risk of loss. The main risk of misuse that we see is that a strategy which passes the checks in this paper is taken as proven to be profitable. Passing them is necessary, but it is not sufficient.

Reproducibility statement

We have tried to make every result in this paper easy to check. First, we provide the source code for all experiments, together with the three input files, the dated out-of-sample forecasts, the trade ledger and machine-readable results, and the replication command rebuilds the 72 fitted fold pipelines. Second, the appendices give the full feature definitions (Appendix A), the fold boundaries and fold-level results (Appendix B), the execution rules (Appendix C), the cost and gating sensitivities (Appendix D), and the commands, software versions, random seeds and input-file hashes needed for replication (Appendix E). Third, we state all datasets, models, hyperparameters and hardware requirements, and we define each metric where it is first used. The whole pipeline runs on an ordinary CPU in a few minutes, and it needs no GPU, database or exchange account.

Appendix A. Feature definitions

Let O, H, L, C, V, Q, N, B denote open, high, low and close prices, base volume, quote volume, number of trades and taker-buy base volume. Let μ_k and σ_k be the trailing k-bar mean and sample standard deviation, and $z_k(X) = (X - \mu_k(X))/\sigma_k(X)$. Every rolling window ends at t. Ratios with a zero denominator are set to missing, and rows with any missing feature are dropped before fitting (in practice only the warm-up rows).

Table A.12: The 30 features.

Group	Count	Definitions
Log returns	6	$\log(C_t/C_{t-k})$ for $k \in \{1, 2, 3, 5, 10, 20\}$
Volatility and location	8	$\sigma_k(\log(C_t/C_{t-1}))$ and $z_k(C_t)$ for $k \in \{5, 10, 20, 60\}$
Bar shape	3	$(H_t - L_t)/C_t$, $(C_t - O_t)/O_t$, $(C_t - L_t)/(H_t - L_t)$
Trading activity	7	B_t/V_t, $\mu_5(B/V)$, Q_t/N_t, $z_{60}(Q/N)$, $z_{60}(N)$, $z_{60}(V)$, $V_t/\mu_{20}(V)$
Technical and calendar	6	RSI(14), MACD histogram, Bollinger position, normalized ATR(14), $\sin(2\pi d/7)$, $\cos(2\pi d/7)$
Total	30	

With $\delta_t = C_t - C_{t-1}$, RSI is $100 - 100/\left[1 + \mu_{14}(\max(\delta, 0))/\mu_{14}(\max(-\delta, 0))\right]$, which uses simple rolling means rather than Wilder's recursive smoothing. The MACD histogram is $M - \mathrm{EMA}_9(M)$ with $M = \mathrm{EMA}_{12}(C) - \mathrm{EMA}_{26}(C)$; exponential means use $\alpha = 2/(k+1)$, start at the first observation and are not bias-adjusted. Bollinger position is $(C_t - \mu_{20}(C))/(2\sigma_{20}(C))$. The true range is $\max\{H_t - L_t, |H_t - C_{t-1}|, |L_t - C_{t-1}|\}$, and normalized ATR is its 14-day simple mean divided by C_t. The weekday d runs from 0 (Monday) to 6. The trading-activity fields summarize the day's trades; they are not order-book snapshots.

Appendix B. Fold boundaries and fold-level accuracy

Table B.13: Fold boundaries and out-of-sample accuracy (%) of the logistic model (L) and inverse persistence (I). "Last label" is the date of the close that defines the last training label; test dates are forecast dates.

Asset	Fold	Train rows	Last label	Test start	Test end	n	L	I
BTC	1	1,000	2020-07-12	2020-07-17	2021-03-23	250	50.0	55.6
BTC	2	1,250	2021-03-19	2021-03-24	2021-11-28	250	50.4	55.6
BTC	3	1,500	2021-11-24	2021-11-29	2022-08-05	250	51.6	50.8
BTC	4	1,750	2022-08-01	2022-08-06	2023-04-12	250	51.6	54.0
BTC	5	2,000	2023-04-08	2023-04-13	2023-12-18	250	54.8	54.8
BTC	6	2,250	2023-12-14	2023-12-19	2024-08-24	250	59.2	54.8
BTC	7	2,500	2024-08-20	2024-08-25	2025-05-01	250	55.2	52.0
BTC	8	2,750	2025-04-27	2025-05-02	2026-01-06	250	52.0	50.0
BTC	9	3,000	2026-01-02	2026-01-07	2026-03-30	83	43.4	42.2

Asset	Fold	Train rows	Last label	Test start	Test end	n	L	I
ETH	1	1,000	2020-07-12	2020-07-17	2021-03-23	250	49.2	52.0
ETH	2	1,250	2021-03-19	2021-03-24	2021-11-28	250	54.4	52.4
ETH	3	1,500	2021-11-24	2021-11-29	2022-08-05	250	48.0	50.8
ETH	4	1,750	2022-08-01	2022-08-06	2023-04-12	250	50.0	54.8
ETH	5	2,000	2023-04-08	2023-04-13	2023-12-18	250	61.6	54.0
ETH	6	2,250	2023-12-14	2023-12-19	2024-08-24	250	54.4	56.0
ETH	7	2,500	2024-08-20	2024-08-25	2025-05-01	250	57.6	54.0
ETH	8	2,750	2025-04-27	2025-05-02	2026-01-06	250	53.2	48.4
ETH	9	3,000	2026-01-02	2026-01-07	2026-03-30	83	42.2	47.0
DOGE	1	1,000	2022-05-30	2022-06-04	2023-02-08	250	55.6	58.8
DOGE	2	1,250	2023-02-04	2023-02-09	2023-10-16	250	58.0	57.6
DOGE	3	1,500	2023-10-12	2023-10-17	2024-06-22	250	52.8	53.6
DOGE	4	1,750	2024-06-18	2024-06-23	2025-02-27	250	52.4	51.2
DOGE	5	2,000	2025-02-23	2025-02-28	2025-11-04	250	48.8	49.2
DOGE	6	2,250	2025-10-31	2025-11-05	2026-03-30	146	49.3	45.2

Training windows start on 16 October 2017 for BTC and ETH and on 3 September 2019 for DOGE, the first dates with a complete feature vector.

Appendix C. Execution engine

The backtest engine iterates over the union of bar timestamps. At each timestamp, for every asset that has a bar, it first resolves exits for open positions in that asset using that bar, then opens positions scheduled for that bar, and finally closes positions whose bar-counted holding period ends at that bar's close. A signal time-stamped within bar i executes at bar $i+1$, at the open adjusted for slippage, $p_{\text{entry}} = O_{i+1}(1 \pm \varsigma)$. Cash and equity follow (1). A long entry debits $q\,p_{\text{entry}} + \varphi$ and a short entry credits $q\,p_{\text{entry}} - \varphi$, where φ is the fee on that leg's own notional. Stops and targets are resolved against each bar's low and high; when one bar touches both, the stop is assumed to fill first. Otherwise a position closes at C_t when its holding period expires. The cash ledger of Section 4.7 is a separate and simpler calculation that uses no barriers.

Appendix D. Cost curve and confidence gating

Table D.14: Logistic book: mean daily net return and annualized net Sharpe ratio at each cost per unit of one-way turnover.

Cost (bps)	Mean net return (bps a day)	Net Sharpe ratio
0	8.70	0.544
1	7.93	0.496
3	6.40	0.400
5	4.87	0.304
7	3.34	0.209
9	1.81	0.113
12	−0.49	−0.031
15	−2.79	−0.174
20	−6.62	−0.414
30	−14.27	−0.892

Table D.15: Confidence gating at 9 bps. An asset keeps its $1/J_t$ allocation only when $|p - 0.5| \geq h$; otherwise that allocation stays in cash. No threshold is selected from these results.

Threshold h	0.00	0.02	0.04	0.06	0.08
Net Sharpe ratio	+0.113	+0.018	−0.080	−0.191	−0.309
Turnover	0.766	0.736	0.642	0.522	0.388

Appendix E. Replication details

The analysis was developed with CPython 3.12.0, NumPy 2.4.4, pandas 3.0.3, SciPy 1.17.1, scikit-learn 1.9.0 and joblib 1.5.3, as pinned in `requirements-research.txt`. After installing them, one command, `python scripts/reproduce_jfds.py`, rebuilds the fold pipelines and forecasts, computes all results and the supplementary diagnostics, and runs the regression and mutation suites. The bootstrap seed is 20,260,401 and the boosted-tree seed is 0. The main outputs are the dated forecasts (`oos_predictions.csv`), the fold manifest, the daily portfolio series, the trade and equity ledgers, `results.json`, `supplementary_diagnostics.json` and `mutation_matrix.json`. Saved model files use Python's joblib format and should be loaded only from trusted sources.

Git can convert line endings on checkout, which changes a file's hash without changing its contents as parsed. Table E.16 therefore gives the SHA-256 hash of each input file both as stored in the repository (LF line endings) and as converted on a Windows checkout (CRLF).

Table E.16: SHA-256 hashes of the input files.

File	Endings	SHA-256
BTCUSDT_daily.csv	LF	5b1b7face0e57c91c0c92a85c0292fed64d64095ab05646a0749c7b3c8b2a70e
	CRLF	aca018e6c7c299b16c691be1e746371f4468c475675899c35bb9f2b109217cfc
ETHUSDT_daily.csv	LF	a2b2aed91296baae8b223dfd3d976c7119d2688c55cd4653d771a827fafbce40
	CRLF	20a0bef67dd349b23c40ae8dd510d0064d4af0def3b687eaf936a95ad444313c
DOGEUSDT_daily.csv	LF	e1523e5b3ab6c9408d7370c217ec5fc79c23d4f922cfc4e9efe6077c92c6f3e4
	CRLF	5eefe6147b3820b39bd7c2c046ef2f0e658e577a23dfcb4d56cfc5dc4a9fce51

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